# Lab 4

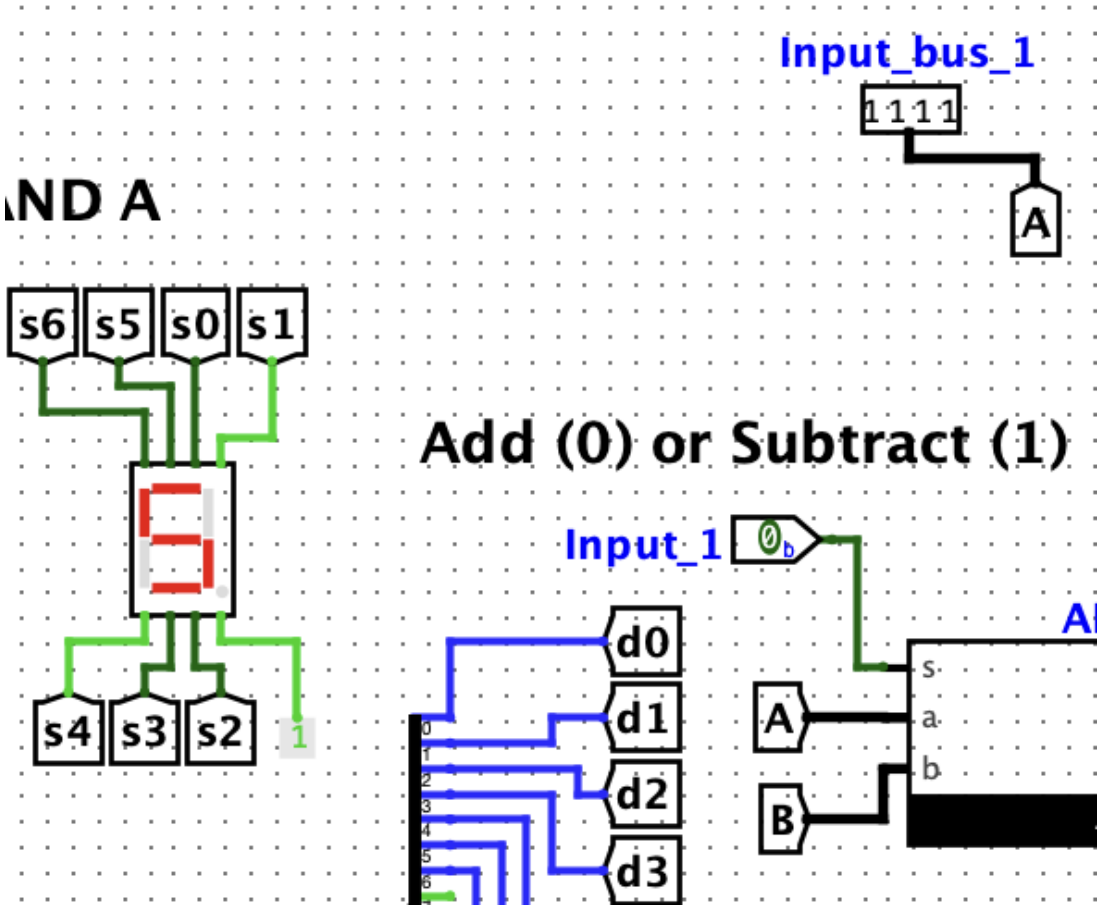
Gerry Glynn 9/29/26

# Inputting binary values for outputs on corresponding row in truth table to create 7_seg_decoder

16 of 16 rows shown

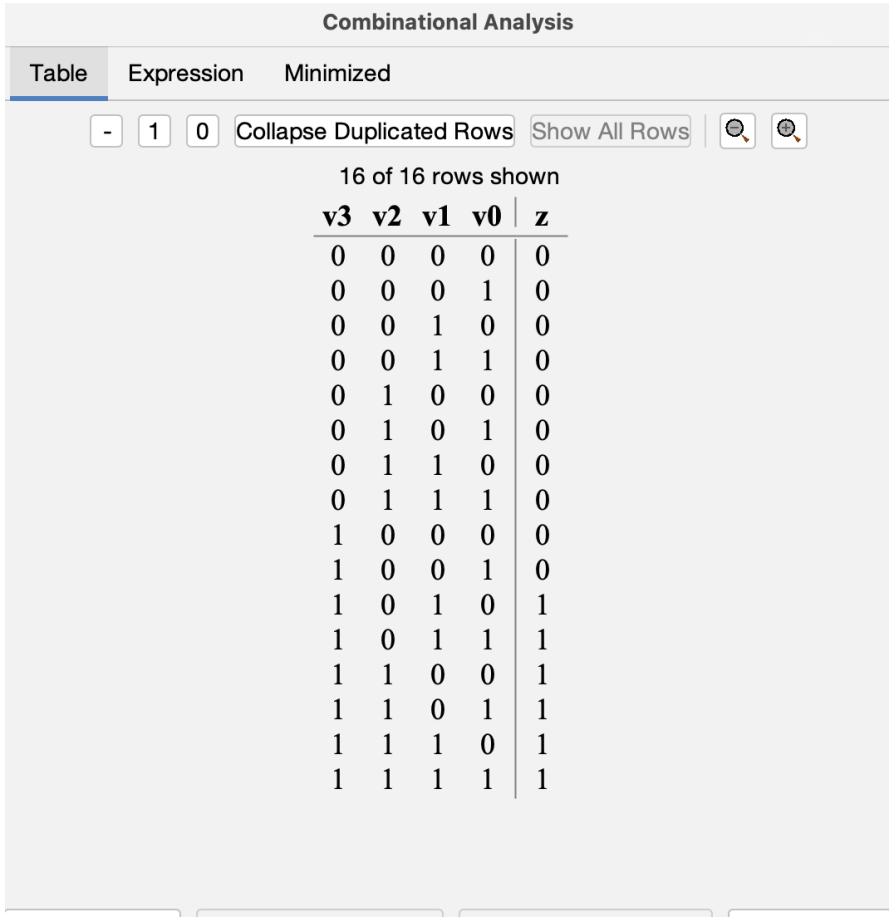
| i_m3 | i_m2 | i_m1 | i_m0 | o_seg[7..0] |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 1 0 0 0 0 0 0 |
| 0 | 0 | 0 | 1 | 1 1 1 1 1 0 0 1 |
| 0 | 0 | 1 | 0 | 1 0 1 0 0 1 0 0 |
| 0 | 0 | 1 | 1 | 1 0 1 1 0 0 0 0 |
| 0 | 1 | 0 | 0 | 1 0 0 1 1 0 0 1 |
| 0 | 1 | 0 | 1 | 1 0 0 1 0 0 1 0 |
| 0 | 1 | 1 | 0 | 1 0 0 0 0 0 1 0 |
| 0 | 1 | 1 | 1 | 1 1 1 1 1 0 0 0 |
| 1 | 0 | 0 | 0 | 1 0 0 0 0 0 0 0 |
| 1 | 0 | 0 | 1 | 1 0 0 1 1 0 0 0 |
| 1 | 0 | 1 | 0 | 1 1 0 0 0 0 0 0 |
| 1 | 0 | 1 | 1 | 1 1 1 1 1 0 0 1 |
| 1 | 1 | 0 | 0 | 1 0 1 0 0 1 0 0 |
| 1 | 1 | 0 | 1 | 1 0 1 1 0 0 0 0 |
| 1 | 1 | 1 | 0 | 1 0 0 1 1 0 0 1 |
| 1 | 1 | 1 | 1 | 1 0 0 1 0 0 1 0 |

# Verifying truth table was done correctly



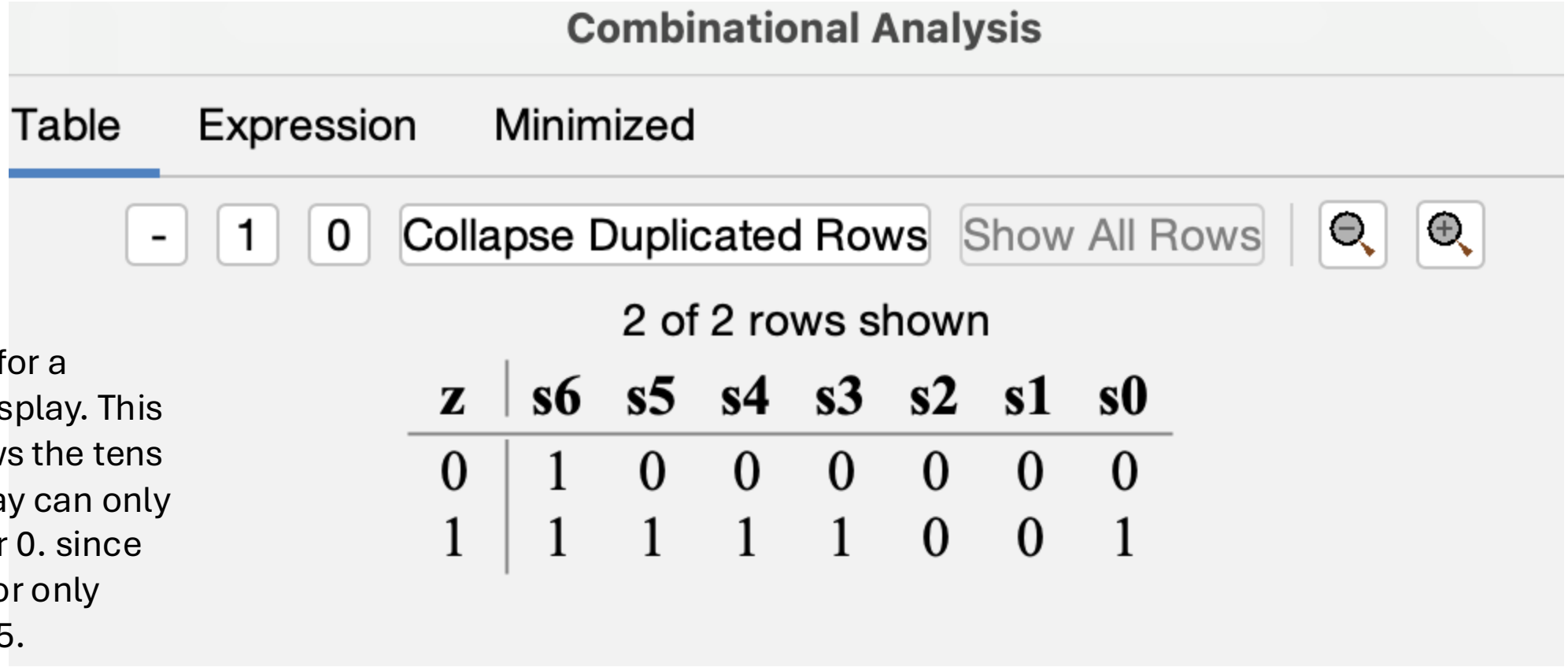
# Comparator truth table built so Circuit B has inputs.

When number is less than 10 z is 0. When number is >= than 10 z is 1.




| v3 | v2 | v1 | v0 | z |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

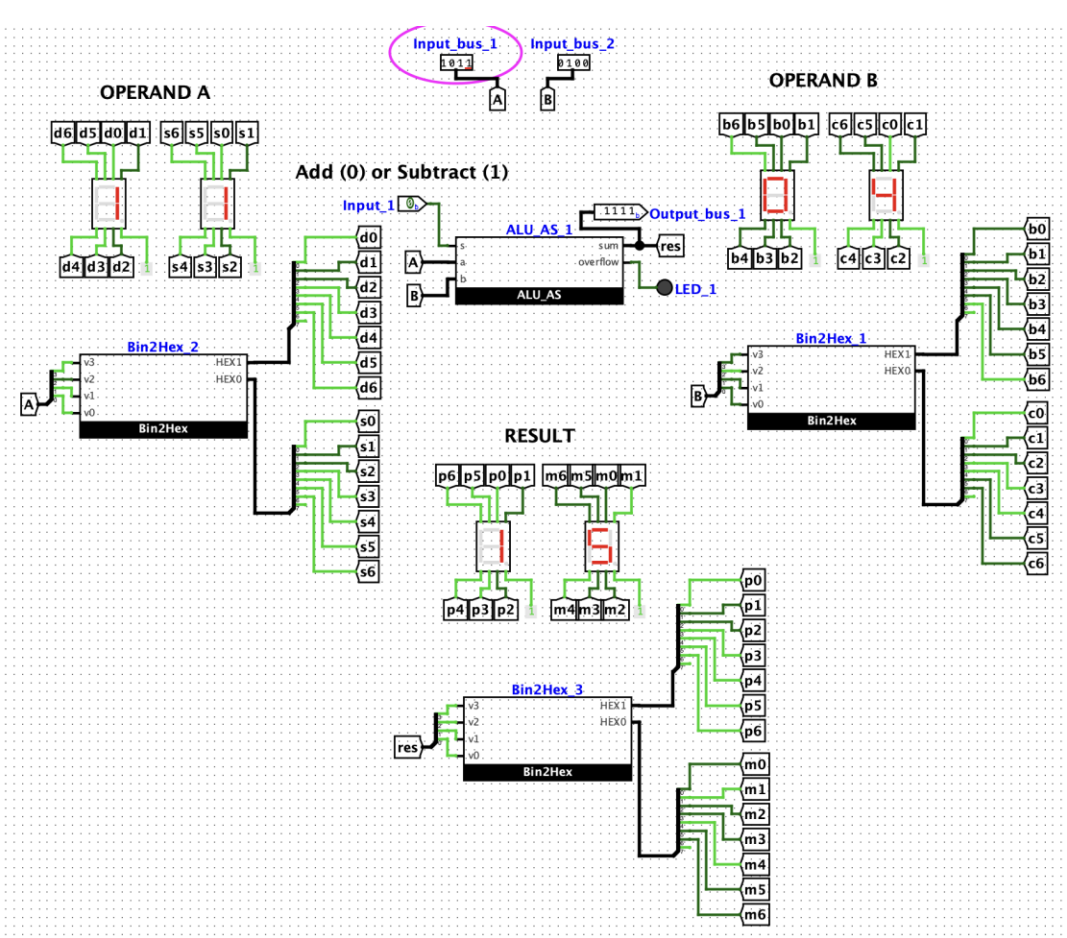
# Truth table created for logic of circuit b

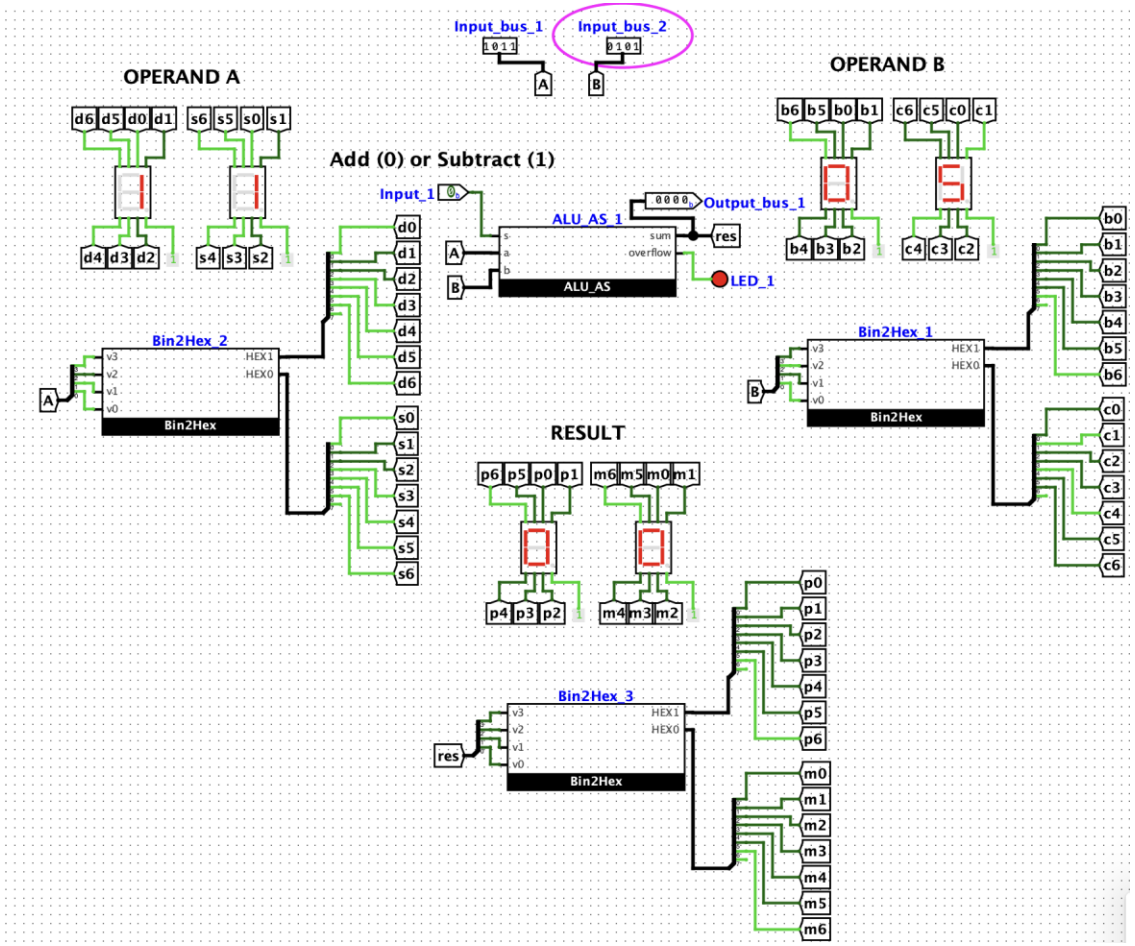



| z | s6 | s5 | s4 | s3 | s2 | s1 | s0 |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |

This table is for a 7segment display. This display shows the tens place. Display can only display a 1 or 0. since our calculator only adds up to 15.

# Circuit created for Lab 4 works but can only adds up to 15



# Video of changing input variables for 7seg_lab4_circuit.

